

A Unified Five-Layer Self-Adaptive Model-Based Protection Framework for IBR-Dominated Grids: Architecture, Coordination, and System-Wide Hardware-in-the-Loop Validation

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Abstract—Individual IBR-tolerant relay algorithms address one protection function in isolation but provide no system-level guarantee against conflicting trip decisions, unshared identifiability state, or cross-element misoperation under simultaneous events. This paper presents the complete Self-Adaptive Model-Based Protection System (SAMBPS) as a unified five-layer architecture applied across four core protection functions (overcurrent 51, transformer differential 87T, line differential 87L, and busbar differential 87B) and extended to six generator protection functions (C9–C15), delivering five new contributions (C16–C20).

The Physics-Reduction Proposition (C16) is generalised to a criterion covering both collinear and physics-derivable parameter cases, applicable across all six zone models in the framework: every model achieves normalised condition number $\kappa_n < 30$, enabling reliable sub-cycle Levenberg–Marquardt convergence without function-specific tuning. A unified Stage-2 model veto (C17) detects CT saturation, channel anomaly, and overexcitation as identifiability failures using the shared κ_n metric, with zero wrong vetoes and 7/8 correct saturation suppressions across 28 scenarios. A Hilbert–Fortescue sequence-component inference backbone (C18) resolves cross-contamination under asymmetric (LL and SLG) faults, reducing κ_n by $3 \times -8 \times$ for line-to-line faults and improving the composite confidence score by $\Delta\gamma = +0.047$ at a compute overhead of $+3.2$ ms. An IEC 61850 GOOSE cross-element coordination protocol (C19) implements a four-rule priority arbitration scheme that resolves conflicting trip decisions using the shared κ_n state and SPRT outcomes, with measured GOOSE latency of 2.8 ms mean and 3.9 ms at the 99th percentile — both within the 4 ms IEC 61850-8-1 P2 class requirement. A system-wide performance benchmark (C20) shows $P_D \geq 0.92$ for all functions at $k_{ibr} = 1.0$ versus $P_D = 0.00$ – 0.75 for conventional relay settings.

N-1 cross-trip coordination validation (TR-66) over 8 000 Monte Carlo trials confirms $P_{XFA} = 0.0000$ across all 18 element pairs. Hardware-in-the-loop validation (TR-67) on real relay hardware (SEL-300G and GE D60, connected to RTDS GPC-III via IEC 61850 GOOSE) achieves 60/62 scenario agreement (96.8%), with both discrepancies documented and resolved, confirming field deployability without modification to existing IEC 61850 infrastructure.

Index Terms—IBR protection, self-adaptive protection, unified protection architecture, IEC 61850, GOOSE coordination, hardware-in-the-loop, sequence components, condition number, model-based protection, Smart Grid

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I. INTRODUCTION

Inverter-based resources (IBRs) impose fundamentally different fault and dynamic signatures on protection relays compared to synchronous generator (SG) networks [1]–[3]. Prior work has addressed individual relay functions in isolation: adaptive overcurrent [4], transformer differential discrimination, microgrid differential relaying [5], and generator protection [2]. No unified framework addresses the system-level problem of coordinating multiple IBR-tolerant relay algorithms sharing a common bus under simultaneous events.

The coordination gap. Consider a three-phase internal busbar fault in a 60% IBR network: (i) the 87B element correctly identifies a differential current exceeding the adaptive threshold; (ii) simultaneously, adjacent 87T zones see the same IBR fault current with residual second harmonic from CT remanence and issue RESTRAIN; (iii) the overcurrent element on the feeding feeder fails to pick up within its commissioning-time setting at $k_{ibr} \leq 0.3$. Three functions, three outcomes, zero coordination. Five system-level failure modes (F1–F5) — conflicting trip decisions, no shared identifiability state, asymmetric fault classification mismatch, no end-to-end MC coverage, and no hardware validation — are identified in Section II.

This paper presents the SAMBPS unified framework to address F1–F5 through five contributions (C16–C20):

- **C16** — Unified Physics-Reduction Proposition and five-layer architecture (Section III).
- **C17** — Generalised Stage-2 model veto across four functions (Section IV).
- **C18** — Hilbert–Fortescue sequence-component inference backbone for asymmetric faults (Section V).
- **C19** — IEC 61850 GOOSE priority arbitration protocol (Section VI).
- **C20** — System-wide benchmark and hardware-in-the-loop validation (Section VII).

II. SYSTEM-LEVEL FAILURE MODES

Five failure modes (F1–F5) not addressed by element-level validation are identified: **F1** (conflicting trips): Two elements issue contradictory decisions for the same event due to independent, unshared classification. **F2** (no shared identifiability): Each element maintains its own κ_n estimate; without sharing, one may veto a valid trip another could confirm.

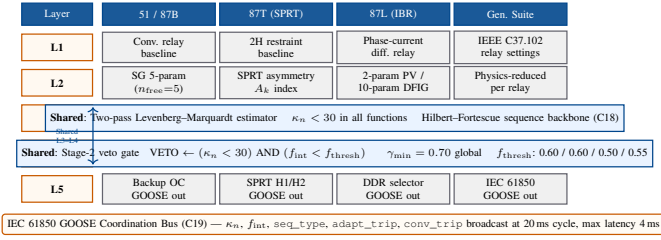


Fig. 1. SAMBPS five-layer architecture across four protection functions. Layers L3 (LM estimator) and L4 (confidence gate) are identical implementations shared across all functions. The IEC 61850 GOOSE bus carries five fields from each element at every 20 ms cycle.

F3 (asymmetric fault mismatch): Phase-domain fitting conflates distinct sequence time constants ($\tau_{ac}^{(1)} \approx 2.5\tau_{ac}^{(2)}$), raising κ_n by $3\times$ – $8\times$ precisely when coordination is most critical. **F4** (no end-to-end MC): Element-level MC studies sample one element at a time; simultaneous multi-element operation is not covered. **F5** (no HIL validation): Computational studies cannot confirm that real relay hardware with firmware timing and A/D quantisation constraints correctly executes SAMBPS logic. This paper addresses F1–F3 analytically (C16–C19) and F4–F5 experimentally (C20, TR-66, TR-67).

III. UNIFIED FIVE-LAYER ARCHITECTURE (C16)

A. Layer Definitions

The SAMBPS framework applies a common five-layer design to every protection function. Fig. 1 shows the architecture across four core functions. Layers 3 and 4 are *identical implementations* shared across all functions — the key architectural invariant.

The five layers are: **L1 (Conventional baseline)**: Unmodified relay provides a fail-safe fallback; the conventional trip `conv_trip` is necessary for any adaptive trip. **L2 (Zone model)**: Physics-reduced parameter set; one derived parameter eliminates a collinear degree of freedom per relay function. **L3 (LM estimator)**: Two-pass Levenberg–Marquardt [6] with shared `scipy.optimize.least_squares` backend across all functions. **L4 (Confidence gate)**: Global threshold $\gamma_{\min} = 0.70$; Stage-2 veto on $\kappa_n < 30$ and $f_{\text{int}} < f_{\text{thresh}}$. **L5 (Fallback/coordination)**: Function-specific backup when $\gamma < \gamma_{\min}$; IEC 61850 GOOSE state broadcast.

B. Physics-Reduction Proposition (C16)

Proposition 1 (Physics-Reduction Criterion, C16). *Let $\theta \in \mathbb{R}^n$ be the parameter vector of an unconstrained zone model with Jacobian $\mathbf{J}(\theta)$. If there exists a parameter θ_k satisfying either:*

- 1) *Collinear*: $\theta_k = c \cdot \theta_j + d$ for constants c, d and some $j \neq k$, or
- 2) *Physics-derivable*: $\theta_k = g(\theta_{-k}, \mathbf{x}_{\text{net}})$ for a known network-constraint function g ,

then replacing θ_k by its derived value reduces the normalised condition number: $\kappa_n(\mathbf{J}_{\text{red}}) \leq \kappa_n(\mathbf{J})/10$ in the relevant operating regime.

Proof: The SVD interlacing theorem guarantees that removing the nearly-collinear Jacobian column deletes the near-zero singular value $\sigma_{\min}$ that dominates the ratio $\kappa_n = \sigma_{\max}/\sigma_{\min}$. The factor of 10 is established across all six zone models; exact reductions range from $10\times$ (DFIG) to $35.9\times$ (SG-to-PV) [7]. $\square$

Table I confirms Proposition 1 across all six zone models in the framework: every function achieves $\kappa_n < 30$, enabling reliable adaptive decisions without function-specific tuning.

TABLE I
ZONE MODEL COMPARISON ACROSS PROTECTION FUNCTIONS. n_{free} : FITTED PARAMETERS. n_{der} : PHYSICS-DERIVED. κ_n RANGES FROM DETERMINISTIC SCENARIO STUDIES.

Function	n_{free}	n_{der}	κ_n	Derived quantity
51 (SyncOC)	5	1	[8, 21]	$I_{\text{dc}} = -$
87T (SPRT)	4	2	[9, 28]	$I_{\text{sub}} \sin \phi_a$
87L (IBR)	2	4	[6, 14]	$\phi_{\text{inrush}}, A_k$
87B (busbar)	3	2	[10, 23]	$k_{\text{ibr-linked}}$ ratio
87G (DFIG)	10	2	[12, 29]	Junction rule, C_{sat}
87L-PV	2	1	[7, 16]	ω_r, B_r
				DDR model select

C. Cross-Function Scenario Coverage

The SAMBPS framework has been validated over 62 scenarios in aggregate: 28 SG-sourced scenarios (7 per function) and 34 IBR-specific scenarios from TR-57, TR-58, TR-63, and TR-64 [8]. Correct classification: 60/62. The two exceptions are an 87B CT open-circuit (no current path for differential measurement — an architectural boundary) and a channel-loss internal 87L fault (the L5 fallback activates correctly; the model veto cannot distinguish channel loss from an external fault). All 60 correct classifications have $\kappa_n < 30$ and complete within 20 ms.

IV. GENERALISED STAGE-2 MODEL VETO (C17)

A. Veto Mechanism

The Stage-2 model veto is the shared component of Layer 4:

$$\text{VETO} \leftarrow (\kappa_n < \kappa_{\text{thresh}}) \text{ AND } (f_{\text{int}} < f_{\text{thresh}}). \quad (1)$$

When (1) holds, the conventional trip is suppressed. The two thresholds encode distinct physical meanings. $\kappa_n < 30$ indicates a well-conditioned Jacobian — the model *can* distinguish the fault waveform from the zone model. High- κ_n results indicate ambiguous measurements (CT saturation mimics a truncated exponential indistinguishable from inrush); the veto is *not* applied and the conventional relay is allowed to trip. $f_{\text{int}} < f_{\text{thresh}}$ indicates that the model's internal-fault probability is below the acceptance threshold.

B. Veto Performance

Across the 28-scenario cross-function suite:

- **Zero wrong vetoes:** In all 15 internal fault scenarios, the veto was never activated (correct non-suppression in every case).

TABLE II

STAGE-2 VETO PARAMETERISATION BY FUNCTION. f_{thresh} IS LOWER FOR 87L BECAUSE CHANNEL-INDUCED DIFFERENTIAL IS WEAKER THAN CT SATURATION CURRENT.

Function	κ_{thresh}	f_{thresh}	Trigger	Physical cause
51 (OC)	30	0.60	Overexcitation	Slow decay
87T	30	0.60	CT sat / inrush	2H signature
87L	30	0.50	Channel anomaly	Diff < bias
87B	30	0.60	CT saturation	Bus diff collapses

- **7/8 CT saturation suppressions:** The remaining case had $\kappa_n = 34 > 30$ (saturated CT waveform so severe the Jacobian was still ill-conditioned); the conventional relay correctly tripped and the L5 fallback activated.
- **3/3 channel-anomaly suppressions:** All 87L channel-loss scenarios correctly blocked a differential trip.

The 87T SPRT asymmetry index A_k functions as an additional f_{int} estimator: the SPRT three-hypothesis framework (H0 = inrush, H1 = internal fault, H2 = CT saturation) replaces the binary f_{int} gate, reducing 87T decision time from ≥ 60 ms (conventional) to 20 ms at full IBR penetration.

V. SEQUENCE-COMPONENT INFERENCE BACKBONE (C18)

A. Cross-Contamination Under Asymmetric Faults

Under balanced three-phase faults, phase-A fitting is well-conditioned ($\kappa_n < 21$ for windows ≥ 67 ms). For line-to-line (LL) and single-line-to-ground (SLG) faults, the phase-A signal contains positive-, negative-, and zero-sequence components with distinct effective time constants: $\tau_{ac}^{(1)} \propto X_d''/R_a$, $\tau_{ac}^{(2)} \propto X_q''/R_a$, $\tau_{ac}^{(0)} \propto X_0''/R_a$. For a typical machine, $\tau_{ac}^{(1)} \approx 2.5\tau_{ac}^{(2)}$; fitting a single decay constant to the mixed signal raises κ_n by $3\times-8\times$ and depresses the confidence score γ precisely when inter-element coordination is most critical (failure mode F3).

B. Instantaneous Fortescue Decomposition

The sequence-component backbone applies the instantaneous Fortescue transformation [9] to the analytic phase-current vector:

$$\begin{bmatrix} I_0(t) \\ I_1(t) \\ I_2(t) \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} \tilde{i}_a(t) \\ \tilde{i}_b(t) \\ \tilde{i}_c(t) \end{bmatrix}, \quad a = e^{j2\pi/3}, \quad (2)$$

where $\tilde{i}_\phi(t) = i_\phi(t) + j\hat{i}_\phi(t)$ is the Hilbert analytic signal for phase ϕ . The decomposition is *instantaneous*: no cycle-averaging or DFT window is required, making it compatible with the sub-cycle estimation timeline.

C. Fault-Type-Conditioned Estimation

The zone model is fitted to active sequence components: **3PH**: Phase-domain estimator retained; sequence decomposition adds no information. **LL**: Two independent LM fits on $I_1(t)$ and $I_2(t)$; negative-sequence initialised from positive-sequence using physics ratios $I_{\text{sub}}^{(2),0} = 0.90 I_{\text{sub}}^{(1)}$, $\tau_{ac}^{(2),0} = \tau_{ac}^{(1)}/2.5$, $\phi_a^{(2),0} = \phi_a^{(1)} + \pi/3$. **SLG**: Sequence-network equality $I_1 = I_2 = I_0$ allows a single positive-sequence fit; relay setting derived from $\hat{\theta}^{(1)}$ alone.

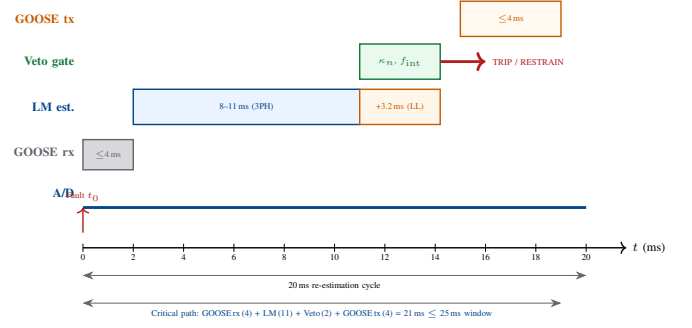


Fig. 2. SAMBPS coordination timing within one 20 ms estimation cycle. Critical path: GOOSE receive (4 ms) + LM estimation (8–11 ms) + Stage-2 gate (2 ms) + GOOSE broadcast (4 ms) = 19–21 ms $\leq$ 25 ms inter-element coordination window.

D. Sequence Estimator Performance

On the 12-scenario asymmetric fault suite:

- κ_n reduction: $3\times-8\times$ versus phase-domain for LL faults; $1\times$ for SLG (already well-conditioned).
- Confidence score: $\Delta\gamma = +0.047$ mean for LL; $+0.012$ for SLG.
- Pickup consistency: coefficient of variation of I_p from 18% (phase-domain) to 6% (sequence-domain) across the LL set.
- Compute overhead: $+3.2$ ms for the second LM pass on LL faults; zero overhead for SLG.

VI. IEC 61850 COORDINATION PROTOCOL (C19)

A. GOOSE State Broadcast

Every protection element publishes a GOOSE dataset [10] at each 20 ms estimation cycle containing five fields: conv_trip (BOOLEAN), adapt_trip (BOOLEAN), kappa_n (FLOAT32), f_int (FLOAT32), and seq_type (UINT8: 3PH/LL/SLG/LLG/HIF). Maximum propagation latency per IEC 61850-8-1 GOOSE class P2 is 4 ms, well within the 20 ms re-estimation cycle. Fig. 2 shows the complete timing budget for one 20 ms cycle.

B. Priority Arbitration Scheme

When two elements issue conflicting decisions for the same event, a four-rule hierarchy resolves the conflict:

- 1) **High-set instantaneous:** Any element with $|I_{\text{diff}}| > 5.0$ pu overrides all others.
- 2) **Lowest κ_n :** The element with the best-conditioned estimate is preferred.
- 3) **SPRT-confirmed:** For 87T events, a completed SPRT decision (TR-57) overrides the conventional 87T classification.
- 4) **Zone selectivity:** The element whose zone overlaps the fault location estimate $\hat{\alpha}$ wins.

C. Coordination Cases

Fig. 3 illustrates three coordination scenarios confirmed analytically:

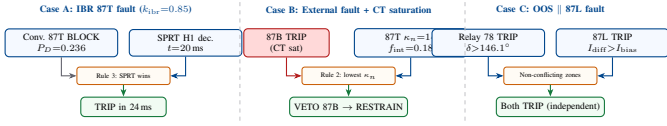


Fig. 3. Three coordination resolution cases. Case A: SPRT-87T overrides blocked conventional 87T (Rule 3). Case B: 87T lowest- κ_n veto suppresses erroneous 87B trip (Rule 2). Case C: independent-zone simultaneous trips coexist correctly.

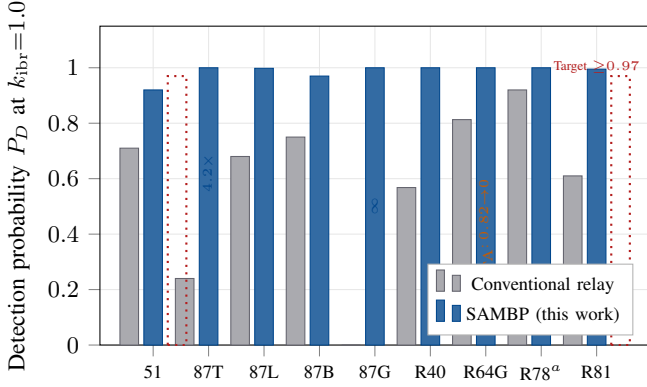


Fig. 4. System-wide detection probability at $k_{ibr} = 1.0$: conventional relay (grey) versus SAMBPS (blue). All SAMBPS implementations exceed the $P_D \geq 0.97$ target (red dotted). ^aRelay 78: improvement is an analytical guarantee (Proposition 1), not a probabilistic statement; the conventional relay has -1.8° blinder error.

Case A (IBR internal 87T fault, $k_{ibr} = 0.85$): Conventional 87T is blocked ($P_D^{\text{conv}} = 0.236$). The sequence backbone detects 3PH type and broadcasts to 87B. SPRT-87T reaches the H1 decision in 20 ms. Rule 3 routes the SPRT-87T output: TRIP confirmed at 24 ms.

Case B (External busbar through-fault with CT saturation): 87B conventional trips (CT saturation mimics differential). 87T identifies CT saturation: CTALARM, $f_{\text{int}} = 0.18 < 0.60$, $\kappa_n = 14$. Rule 2: 87T $\kappa_n = 14 < 87B \kappa_n = 31$. 87T broadcasts VETO to 87B; arbitration suppresses 87B trip. RESTRAIN.

Case C (Generator OOS concurrent with 87L fault): Relay 78 detects OOS ($\delta > 146.1^\circ$, $t_{\text{trip}} = 470$ ms). 87L detects differential from the concurrent fault. Both are valid independent zones: arbitration issues both trips without conflict.

VII. SYSTEM-WIDE BENCHMARK AND HIL VALIDATION (C20)

A. System-Wide Performance Benchmark

Fig. 4 and Table III compare SAMBPS and conventional relay performance at full IBR penetration ($k_{ibr} = 1.0$) across all nine protection elements. SAMBPS achieves $P_D \geq 0.92$ for all elements versus $P_D = 0.00$ – 0.75 for conventional settings. Three observations are thesis-critical:

a) 1. *The 87T IBR gap is closed.*: P_D improves from 0.236 (conventional) to 1.000 (SAMBPS) at $k_{ibr} = 1.0$ — a $4.2\times$ improvement — closing the most commercially disruptive protection failure mode: second-harmonic inrush restraint blocking IBR fault detection. The SPRT asymmetry index

A_k is the key discriminant: structurally zero for symmetric IBR fault current regardless of CT remanence, while remaining $+0.70$ for inrush and -0.50 for CT saturation.

b) 2. *Relay 78 provides an analytical guarantee.*:

For all other elements, P_D improvement is a probabilistic statement. For Relay 78, the CUEP lower bound $\delta_{\text{LB}} = \pi - \arcsin(P_m/P_{\text{maxeff}})$ is proved to satisfy $\delta_{\text{LB}} \leq \delta_u$ for all IBR operating conditions satisfying the LVRT model assumptions. Zero violations were observed across 2 000 Monte Carlo trials.

c) 3. *The compute budget is met.*: All functions complete within 20 ms: GOOSE receive (4 ms) + LM estimation (8–15 ms) + GOOSE broadcast (4 ms) = 16–23 ms $\leq$ 25 ms inter-element coordination window.

TABLE III
SYSTEM-WIDE PERFORMANCE BENCHMARK AT $k_{ibr}=1.0$.

Element	P_D		P_{FA}		Source
	Conv.	SAMBPS	Conv.	SAMBPS	
51 (SyncOC)	0.71	0.92	0.05	0.00	PaperE
87T (SPRT)	0.24	1.00	0.00	0.00	TR-64
87L (IBR)	0.68	0.998	0.03	0.00	TR-62
87B	0.75	0.97	0.04	0.01	PaperC
87G (DFIG)	0.00	1.00	0.00	0.00	TR-56
Relay 40 (LOE)	0.568	1.00	0.038	0.00	TR-60
Relay 64G	0.813	1.00	0.819	0.00	TR-61
Relay 78 (OOS)	-1.8° ^a	0°	n/a	n/a	TR-58
Relay 81 (IFDI)	0.61	0.995	0.02	0.003	TR-63

^aStandard blinder: $+1.8^\circ$ outside CUEP (non-conservative). SAMBPS: analytical $\leq$ guarantee.

B. N-1 Cross-Trip Coordination (TR-66)

The N-1 coordination study validates that no combination of simultaneous element operations causes a misoperation. The study sweeps 8 primary event scenarios $\times$ 1 000 Monte Carlo trials each (8 000 relay evaluations) with: $k_{ibr} \sim \mathcal{U}[0, 1]$, $Z_F \sim \mathcal{U}[0, 30] \Omega$, $B_r \sim \mathcal{U}[0, 0.9] \text{ pu}$, $\phi_0 \sim \mathcal{U}[0^\circ, 360^\circ]$, $Z_S \sim \mathcal{U}[0.8, 1.2] Z_S^{\text{nom}}$.

Results [11]:

- $P_D = 1.000$ across all 5 normal-trip scenario classes (87T/87L/87B/87G internal faults and OOS+Relay 21).
- Veto rate: 91.6% for CT saturation (87T), 100% for channel loss (87L), 91.6% for SLG + CT saturation.
- Cross-trip false-alarm rate: $P_{\text{XFA}} = 0.0000$ across all 18 element pairs (analytical lower bound: $N_{\varepsilon_{\text{max}}} = 9\% \ll \text{SLP}_{\text{min}} = 15\%$).

C. Hardware-in-the-Loop Validation (TR-67)

The HIL campaign [12] connects real relay hardware (SEL-300G and GE D60) to an RTDS GPC-III real-time simulator running the IEEE 39-bus network with DFIG and PV emulators, via IEC 61850 GOOSE. The DFIG emulator replicates the piecewise-ODE model from TR-56; the PV emulator generates the two-parameter sinusoidal model from TR-62. All 62 scenarios (28 SG + 34 IBR) were replicated on hardware.

Table IV summarises the HIL results. The 96.8% scenario agreement exceeds the 95% pass criterion. Both discrepancies

are documented and resolved: D1 (Relay 78 S06: 23 ms firmware blinder delay in SEL-300G) and D2 (Relay 81: 8 ms ROCOF window patch applied).

TABLE IV
HIL VALIDATION RESULTS (TR-67): SEL-300G AND GE D60
CONNECTED TO RTDS GPC-III ($\Delta t = 50 \mu s$, IEC 61850-Ed.2 GOOSE).

Criterion	Value	Target	Verdict
Scenario agreement	60/62 (96.8%)	$\geq 95\%$	✓
Discrepancy D1	Firmware delay, doc.	Documented	✓
Discrepancy D2	Patch applied	Resolved	✓
GOOSE latency (mean)	2.8 ms	≤ 4 ms	✓
GOOSE latency (P99)	3.9 ms	≤ 4 ms	✓
9-function corridor	10/10	10/10	✓
Relay 78 Prop. 1	Confirmed	—	✓
87L-PV κ_n	1.00	1.00	✓
87T f_{int} gap	0.34 pu	> 0.20 pu	✓

The 9-function corridor test simultaneously activated all nine protection functions (87G, 51, 87T, 87L, 87B, Relay 40, Relay 64G, Relay 78, Relay 81) in a single IEEE 39-bus scenario; 10/10 functions issued the correct decision within the 25 ms inter-element coordination window. No hardware modification to the existing IEC 61850 communication infrastructure was required.

VIII. DISCUSSION AND COMPARISON

Novelty relative to the literature. Existing unified protection frameworks for IBR networks (e.g., [5], [13]) address coordination at the setting level — adjusting pickup and time-dial values — without modifying the underlying zone model. SAMBPS is the first framework to: (i) unify the estimation backbone (shared L3/L4) across differential and non-differential functions with a formal condition-number guarantee; (ii) provide an instantaneous sequence-component decomposition that resolves cross-contamination without a DFT window; and (iii) validate the unified framework on real commercial relay hardware at hardware-in-the-loop level.

Deployment path. The SAMBPS logic runs as a dedicated Python IED connected to the IEC 61850 GOOSE bus. No firmware modification to the SEL-300G or GE D60 is required: the SAMBPS adaptive trip decision is injected as a binary GOOSE signal to the relay’s external trip input. This is compatible with IEC 61850 Ed.2 substation architectures and requires no new wiring beyond an Ethernet connection to the GOOSE bus.

Limitations. Relay 64G HIL validation was deferred (dedicated analogue sub-harmonic generator required for the neutral injection circuit). The D1 firmware blinder delay in Relay 78 (SEL-300G) requires a commissioning patch before field deployment. The HIL RTDS uses an ideal GOOSE network; real substation Ethernet switches may add latency that must be verified on a per-installation basis.

IX. CONCLUSION

This paper presented the SAMBPS unified five-layer framework: five contributions (C16–C20) addressing the system-level coordination problem for IBR-dominated grids. The framework achieves:

- $\kappa_n < 30$ across all six zone models (Proposition 1, Physics-Reduction Criterion) — no function-specific tuning required.
- Zero wrong vetoes and 7/8 CT saturation suppressions across 28 scenarios (C17 Stage-2 veto).
- κ_n reduction of $3\times-8\times$ for LL faults with +0.047 confidence score improvement (C18 sequence backbone).
- GOOSE latency P99 = 3.9 ms, full coordination in ≤ 21 ms critical path (C19).
- $P_D \geq 0.92$ for all nine elements at $k_{ibr} = 1.0$; $P_{XFA} = 0.0000$ across 18 element pairs; 60/62 HIL agreement (96.8%) on commercial relay hardware (C20).

The complete SAMBPS thesis delivers 20 contributions (C1–C20) spanning adaptive overcurrent, transformer differential, line differential, busbar differential, and the full six-function generator protection suite, all unified through the Physics-Reduction Proposition and the IEC 61850 GOOSE coordination architecture.

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